

**ASE2910 Applied Linear Algebra / AUS2910 Fundamental Math for AI
Homework #3**

- 1) *Map on a Napkin.* You choose three non-orthogonal direction hints in $\mathbb{R}^3$,

$$u_1 = (1, 1, 0), \quad u_2 = (0, 1, 1), \quad u_3 = (1, 0, 1),$$

and wish to construct an orthonormal world frame q_1, q_2, q_3 via Gram-Schmidt algorithm. Then, express the landmark

$$p = (6, 5, 4)$$

in the q -coordinates, *i.e.*, compute $(\langle q_1, p \rangle, \langle q_2, p \rangle, \langle q_3, p \rangle)$. Provide all steps symbolically (with exact radicals), and verify orthonormality explicitly.

- 2) *Matrix equality.* Let A and B be two $m \times n$ matrices. Under each of the assumptions below, determine whether $A = B$ must always hold, or whether $A = B$ holds only sometimes.

- a) Suppose $Ax = Bx$ holds for all n -vectors x .
- b) Suppose $Ax = Bx$ for some nonzero n -vector x .

- 3) *Transformation by orthonormal vectors.* Suppose that the columns of $T \in \mathbb{R}^{n \times k}$ with $k \leq n$ are orthonormal. Show that the transformation $x \mapsto Tx$ satisfies the following properties.

- a) The transformation is *isometric*, *i.e.*, it preserves *distance* between vectors,

$$\|Tx - Ty\| = \|x - y\|, \quad \forall x, y.$$

- b) The transformation is *conformal*, *i.e.*, it preserves *angle* between vectors,

$$\angle(Tx, Ty) = \angle(x, y), \quad \forall x, y.$$

4) *Rotation matrices.* Consider a matrix A that describes a rotation by θ , that is,

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}_y = \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x$$

- a) Explain why $\|y\| = \|x\|$ for any x and θ .
- b) Show that the columns of A are orthonormal vectors.
- c) Construct a matrix that describes a rotation by $-\theta$?
- d) What is A^T ? Is it equal to what you obtained from above?
- e) Consider a vector x , and suppose that we compute $y = Ax$, and then subsequently compute $z = A^T y$. What is z ?
- f) What is $A + A^T$? What does it do? Justify your answer by drawing a picture on a plane to illustrate x , Ax , $A^T x$, and $(A + A^T)x$

5) *Quadratic form.* Suppose P is an $n \times n$ matrix. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined as $f(x) = x^T P x$ is called a *quadratic form*, and generalizes the idea of a quadratic function of a scalar variable, px^2 . The matrix P is called the coefficient matrix of the quadratic form.

- a) Show that $f(x) = \sum_{i,j} P_{ij} x_i x_j$. In words: $f(x)$ is the weighted sum of all products of two components of x , with weights given by the entries of P .
- b) Show that for any x , we also have $f(x) = x^T P^T x$. In other words, the quadratic form associated with the transpose matrix is the same function.
- c) Show that f can be expressed as $f(x) = x^T P^s x$, where $P^s = (1/2)(P + P^T)$ is the symmetric part of P . The matrix P^s is symmetric. So any quadratic form can be expressed as one with a coefficient matrix that is symmetric.
- d) Express $f(x) = -2x_1^2 + 4x_1 x_2 + 2x_2^2$ in the form $f(x) = x^T P x$ with P a symmetric 2×2 matrix.
- e) Suppose that A is an $m \times n$ matrix and b is an m -vector. Show that $\|Ax - b\|^2 = x^T P x + q^T x + r$ for a suitable $n \times n$ symmetric matrix P , n -vector q , and constant r . (Give P , q , and r .) In words: The norm squared of an affine function of x can be expressed as the sum of a quadratic form and an affine function.

6) *VMLS Exercises.*

- a) **5.6** *Gram–Schmidt algorithm.*
- b) **5.8** *Early termination of Gram–Schmidt algorithm.*
- c) **6.3** *Block matrix.*
- d) **6.8** *Cash flow to bank account balance.*
- e) **6.10** *Resource requirements.*
- f) **6.12** *Skew-symmetric matrices.*
- g) **6.14** *Norm of matrix-vector product.*
- h) **6.17** *Stacked matrix.*
- i) **7.1** *Projection on a line.*
- j) **7.2** *3-D rotation.*
- k) **7.3** *Trimming a vector.*
- l) **7.4** *Down-sampling and up-conversion.*
- m) **8.3** *Cross product.*
- n) **8.8** *Interpolation of rational functions.*
- o) **8.11** *Location from range measurements.*